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Optimal design of a photovoltaic–diesel–battery hybrid renewable energy system for sustainable off-grid electrification of a village in Ethiopia

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Abstract

Rural electrification in developing regions remains a major challenge due to the absence of grid infrastructure and high costs of energy distribution. This study presents the design, simulation, and optimization of a hybrid photovoltaic (PV)–diesel–battery system to supply reliable electricity to an off-grid village in the Jawi region of Ethiopia. With an annual load demand of 153,396.36 kWh and average solar radiation of 1211.8 kWh/m²/year, a hybrid configuration comprising an 81 kW PV array, 25 kW diesel generator, 45 kW converter, and 200 batteries was modeled using PVsyst and HOMER software. Various system configurations were evaluated based on net present cost (NPC), cost of energy (COE), fuel consumption, and CO₂ emissions. The optimized hybrid system achieved a net present cost of 7,284,233 Birr, a cost of energy of 4.468 Birr/kWh, annual diesel consumption of 9,377 L, and CO₂ emissions of 24,693 kg, with a renewable contribution of 84%. Compared to the diesel-only configuration, the hybrid system significantly reduced both fuel usage and emissions while improving reliability and long-term sustainability. Sensitivity analyses confirmed system robustness against variations in fuel price and PV cost. The results demonstrate that a PV–diesel–battery hybrid system offers a technically feasible, economically viable, and environmentally sustainable solution for off-grid electrification in rural Ethiopia.

Keywords Hybrid energy system, Photovoltaic, Diesel generator, HOMER simulation, Off-grid electrification, Ethiopia

Introduction

Rural electrification continues to pose a major challenge in developing countries where grid extension is economically unfeasible and infrastructure remains underdeveloped. Reliable access to electricity is essential for social and economic growth, supporting critical sectors such as healthcare, education, agriculture, and communication. To address these challenges, Hybrid Renewable Power Generation Systems (HRPGS) have emerged

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as an effective alternative for providing sustainable and decentralized energy solutions [1]. These systems combine renewable and conventional energy sources to ensure a stable, cost-effective, and environmentally friendly power supply, especially in remote and off-grid regions. Among various renewable technologies, solar and wind energy are the most widely used in hybrid configurations. Solar photovoltaic (PV) technology, in particular, offers high scalability, low maintenance, and the ability to provide direct electricity generation from sunlight [2]. A standard PV system consists of solar arrays, power conditioning units (inverters and regulators), and battery storage, which together ensure continuous and reliable energy delivery. The system design typically involves assessing solar potential, estimating load demand, and optimizing system components such as inverters and batteries for maximum efficiency. This modular and flexible nature makes PV technology particularly suitable for decentralized or distributed power generation, reducing transmission losses and improving energy accessibility in isolated communities. Recent studies have further emphasized the relevance of hybrid renewable systems in Sub-Saharan Africa, demonstrating the techno-economic feasibility of 100% renewable configurations integrating PV, biomass, and pumped-hydro storage to support rural communities [3].

Literature survey

This section presents a detailed review of previous studies on hybrid renewable energy systems, including PV, wind, biomass, diesel, and hydrogen-integrated configurations. The purpose is to contextualize the present work within existing research and highlight key advancements, limitations, and technological trends relevant to rural electrification. All content that follows in this section builds upon the introductory motivation without altering it.

Several researchers have explored hybrid and standalone renewable systems for rural and off-grid applications. Early investigations assessed the feasibility of centralized charging systems for solar PV lanterns as an alternative to kerosene lamps for rural lighting [4]. The study demonstrated that PV-based charging networks not only provided reliable illumination but also generated local employment and promoted economic sustainability. Another study focused on developing a hybrid air-electricity storage system integrated with PV generation, where the coordination of PV panels, battery storage, and air compression demonstrated operational feasibility and improved overall system stability [5]. Similarly, a Graph Theoretic Modeling (GTM) approach was used to develop a model for standalone PV systems applied to solar ventilators and water pumping, highlighting its potential for integration into simulation-based design software despite the absence of concurrent equation-solving capabilities [6]. Socioeconomic aspects were also examined through case studies on rooftop PV installations, revealing that the high initial investment, limited subsidies, and inadequate financial support were the primary barriers to solar adoption in Austria [7]. Dust accumulation and environmental conditions are additional factors influencing PV efficiency; studies on wind velocity and airborne dust concentration demonstrated that higher wind speeds exacerbate performance degradation in PV cells due to increased dust particle interactions [8]. Building upon such research, recent optimization studies in remote regions of Nigeria and Cameroon have demonstrated that hybrid PV–wind–battery or PV–wind–diesel systems can achieve significant cost savings and emission reductions compared to

existing electrification methods, underscoring their suitability for off-grid communities [9]–[10].

Research on biomass–PV hybrid systems proposed methodologies for determining cost-effective sizing ratios between primary and backup energy sources. Such configurations were found to be both economically viable and environmentally sustainable, particularly for rural regions abundant in biomass resources [11]. Optimization models for photovoltaic–diesel hybrid systems have also been developed, minimizing energy costs while satisfying system autonomy constraints. These analyses revealed that hybrid systems could reduce required battery capacity by nearly 50% compared to standalone PV systems [12]. Operational strategies for PV–diesel–battery configurations have been proposed to enhance cost efficiency by optimizing generator usage and accounting for battery degradation [13]. Simulation-based investigations on small-scale PV–diesel systems for marine applications demonstrated that hybridization improves reliability under fluctuating solar conditions compared to standalone PV systems [14]. Further studies confirmed that incorporating energy storage in renewable-based hybrid systems enhances autonomy, ensures reliable power delivery during variable weather conditions, and achieves cost reductions compared to diesel-only systems [15]. Hybrid configurations also provide improved operational stability, lower maintenance needs, and reduced downtime, while utilizing locally available renewable resources that contribute to lower emissions and energy independence [16]. Recent works have expanded these concepts by evaluating biomass-based power technologies, hydrogen energy storage, and H₂-integrated hybrid configurations, demonstrating their growing relevance for sustainable rural electrification in Sub-Saharan Africa [17–20].

Despite the extensive literature on hybrid energy systems, limited research has focused on comprehensive techno-economic optimization of PV–diesel–battery configurations tailored to the Ethiopian context, where solar resources are abundant but electrification rates in rural areas remain low. Previous studies have not sufficiently addressed site-specific system sizing, cost optimization, and environmental analysis under realistic rural load conditions.

The present study aims to design, model, and optimize a PV–Diesel–Battery hybrid power system for rural electrification in the Jawi region of Ethiopia using HOMER and PVsyst software. The specific objectives are:

- To evaluate the solar energy potential and energy demand of the selected rural site.
- To design an optimal hybrid configuration that minimizes system cost and fuel dependency.
- To perform sensitivity analysis for varying PV prices and fuel costs.
- To assess environmental benefits through CO₂ emission reduction.

The novelty of this work lies in its integrated simulation and optimization framework combining HOMER and PVsyst for the Ethiopian rural context, enabling precise evaluation of both technical and economic performance. The proposed model serves as a replicable blueprint for designing hybrid renewable energy systems in other off-grid regions of Sub-Saharan Africa.

Contributions and novelty of the study

This study provides several contributions beyond conventional HOMER-based simulations. First, it applies an integrated modeling workflow that combines detailed PV module performance characterization using PVsyst with techno-economic optimization in HOMER, ensuring that temperature-corrected efficiencies, inverter behavior, and real irradiance conditions are incorporated into the final system sizing. Second, the work develops a site-specific modeling framework tailored to the rural context of Jawi, including locally measured solar parameters, realistic load patterns, and regionally relevant cost structures, which have not been systematically analyzed in prior Ethiopian studies. Third, the study evaluates system robustness through sensitivity analyses anchored in local market uncertainties, providing insights directly usable for rural electrification planning. Together, these elements constitute a comprehensive framework that extends beyond standard simulation practices and provides actionable design guidance for off-grid communities in Ethiopia.

The remainder of this article is organized as follows: Sect. "[Methodology](#)" presents the methodology adopted to carry out the study. The results and discussion are presented in Sect. "[Results and discussion](#)". The conclusions end the paper in Sect. "[Conclusion](#)".

Methodology

This study adopts a hybrid photovoltaic–diesel–battery configuration because it offers the most practical, cost-effective, and reliable solution for rural electrification in remote Ethiopian regions where grid access is unavailable. The methodology combines standard design practices commonly used for standalone hybrid systems with a unique dual-software simulation approach using both HOMER and PVsyst, enabling a more comprehensive techno-economic and performance analysis than conventional single-platform studies. HOMER was selected for optimization and economic evaluation due to its proven capability in hybrid system planning, while PVsyst was used for detailed PV module performance modeling and loss analysis. This integrated approach allows accurate sizing, realistic performance estimation, and robust sensitivity analysis, making the methodology more precise and context-specific compared to typical standalone PV or PV–Diesel assessments.

In this study, the methodological framework combines both well-established hybrid system design procedures and newly integrated analytical approaches. Standard steps such as solar radiation estimation using the Angstrom–Prescott model, PV sizing, battery autonomy calculations, and selection of system voltage follow widely accepted practices, which are described briefly and supported with appropriate citations. In contrast, the unique aspects of this work—including the dual-software simulation strategy using HOMER for techno-economic optimization and PVsyst for detailed PV performance modeling, as well as the site-specific sensitivity analysis tailored to Ethiopian rural conditions—are presented in greater detail. This structure ensures that readers can clearly differentiate traditional procedures from the novel methodological contributions of this study.

Site description

The case study focuses on Jawi (Buni Jirra), located in the Agew Awi Zone of Ethiopia's Amhara Region, at latitude 10.624°N and longitude 36.37°E. The area's average daily

solar irradiation is 5.93 kWh/m²/day, with maximum values between February and April and minimum values during the rainy months (June–July). This variation makes it an ideal site for testing a hybrid solar–diesel configuration that can accommodate seasonal changes in sunlight availability.

Solar radiation Estimation and panel orientation

In Ethiopia, solar radiation data are often unavailable; therefore, the Angstrom–Prescott correlation was employed to estimate monthly average solar radiation from sunshine duration [21]. The monthly average daily global radiation on a horizontal surface (H) is given by:

$$\frac{\bar{H}}{\bar{H}_o} = a + b \frac{\bar{n}_s}{\bar{N}_s} \quad (1)$$

where: $\bar{H}$ - Monthly average daily radiation on horizontal surface.

$\bar{H}_o$ - Monthly average radiation outside of the atmosphere on the horizontal surface for the same location.

$\bar{n}_s$ - Monthly average daily hours of bright sunshine.

$\bar{N}_s$ - Monthly average of the maximum possible daily hours of bright sunshine.

a and b are regression coefficients which were determined from Klein relationship.

For Jawi, the optimal tilt angle was selected as 10.624°, corresponding to the site's latitude. Panels were oriented due south to maximize annual solar gain in the Northern Hemisphere.

Figures 1, 2 and 3 illustrate the monthly variations of solar radiation. The radiation is highest from February to April (6.3–6.7 kWh/m²/day) and lowest in June–July (around 5.06 kWh/m²/day). The tilted surface performs better in winter, while the horizontal surface slightly outperforms it during summer. These patterns are critical for determining the PV array's orientation and optimizing energy production throughout the year.

Determination of available sunlight

The design is based on the worst month solar availability, ensuring that the system can operate reliably throughout the year. The site's average annual solar energy of 5.934 kWh/m²/day was used as the baseline input for PVsyst and HOMER simulations.

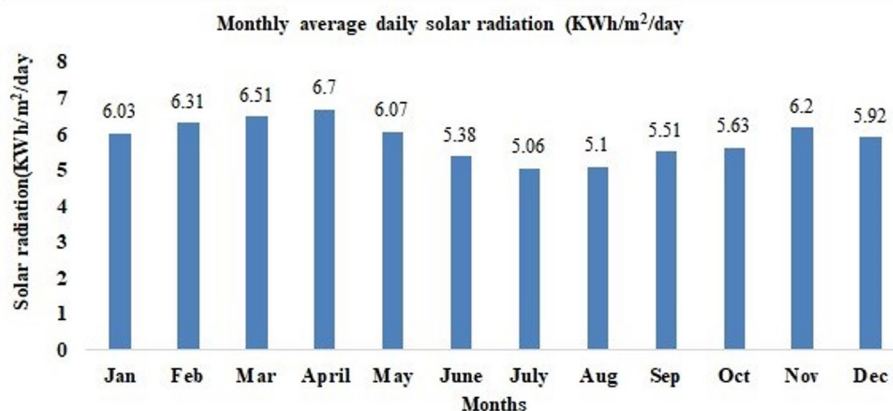


Fig. 1 Monthly average daily solar radiation (KWh/m²/day) without tilt angle

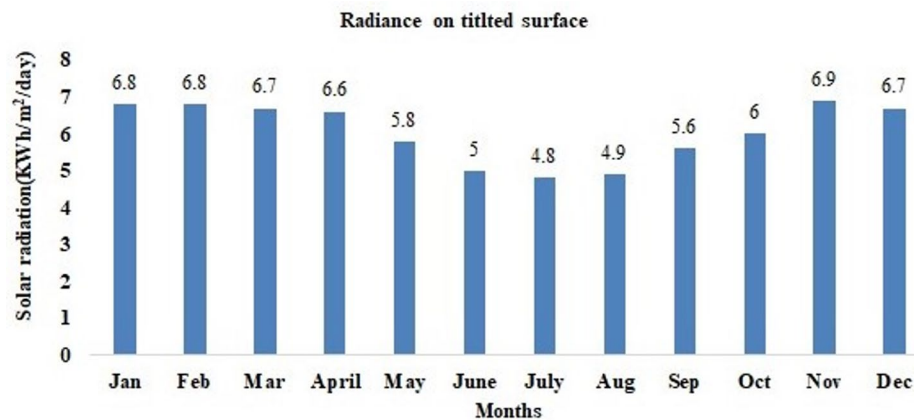


Fig. 2 Monthly average daily solar radiation (KWh/m²/day) using a 10.624° tilt angle

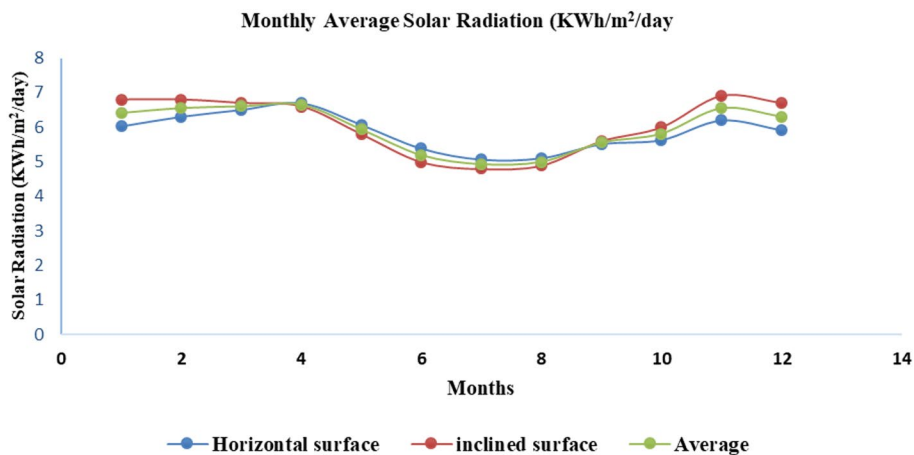


Fig. 3 Comparison of solar radiation on horizontal and tilted surface of PV panels

Hybrid system mathematical modeling and configuration

The most suitable configuration for rural power generation combines solar PV and diesel generator systems with battery storage and converters. All components are interconnected to ensure system efficiency and continuous power availability. Figure 4 shows the schematic layout of the hybrid system, consisting of PV panels, a diesel generator, converter/inverter, and a battery bank. The controller manages the energy flow to maintain stability and efficiency.

Operational sequence:

PV panels generate DC electricity, which is converted to AC for load use.

Excess energy charges the batteries.

When solar output is insufficient, stored energy from the batteries supplies the load.

If both solar and battery energy are unavailable, the diesel generator provides backup power. In this way, the hybrid system ensures continuous electricity using solar energy when available, batteries as support, and the diesel generator as the last backup [22].

Component design and sizing

The system voltage is chosen based on how much power the system needs to handle. In general, higher daily loads require higher system voltages. For standalone solar PV

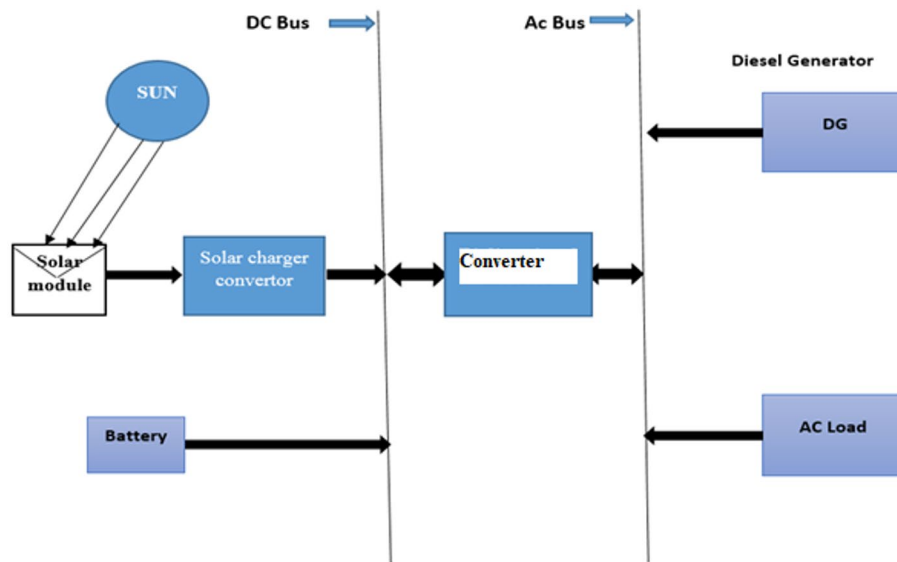


Fig. 4 Block diagram of hybrid power generation system

systems, the choice also depends on the types of inverters available, since they must match the system voltage. When the connected loads need AC power, the DC system voltage should be selected after checking the inverter specifications. In this case, because the total AC load is more than 5000 W, a 48 V DC system voltage has been selected [23, 24].

Component sizing ensures optimal balance between performance and cost.

PV array The number of modules was calculated using.

$$\text{Number of modules} = \text{System power} / P_{\max} \quad (2)$$

The system includes an 81 kW PV array.

Battery bank Battery Capacity was estimated using.

$$C_X = \frac{N_C * E_L}{DOD_{MAX} * V_{SYSTEM} * \eta_{OUT}} \quad (3)$$

where,

C_X = required battery capacity, N_C = number of days autonomy, E_L = estimated load energy in WH.

DOD_{MAX} = maximum depth of discharge, η_{OUT} = Battery efficiency.

A total of 200 batteries were selected for energy autonomy during low-sun months.

Inverter/Converter: The inverter capacity was set to 45 kW to handle AC/DC conversions efficiently.

Diesel generator: A 25 kW diesel generator was selected as a backup power source, operating under a Load Following (LF) control strategy.

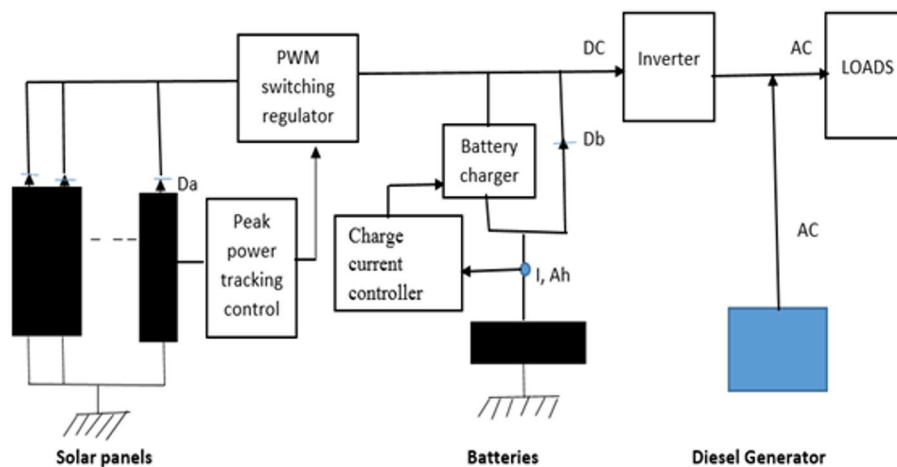


Fig. 5 Total PV System Components with diesel generator

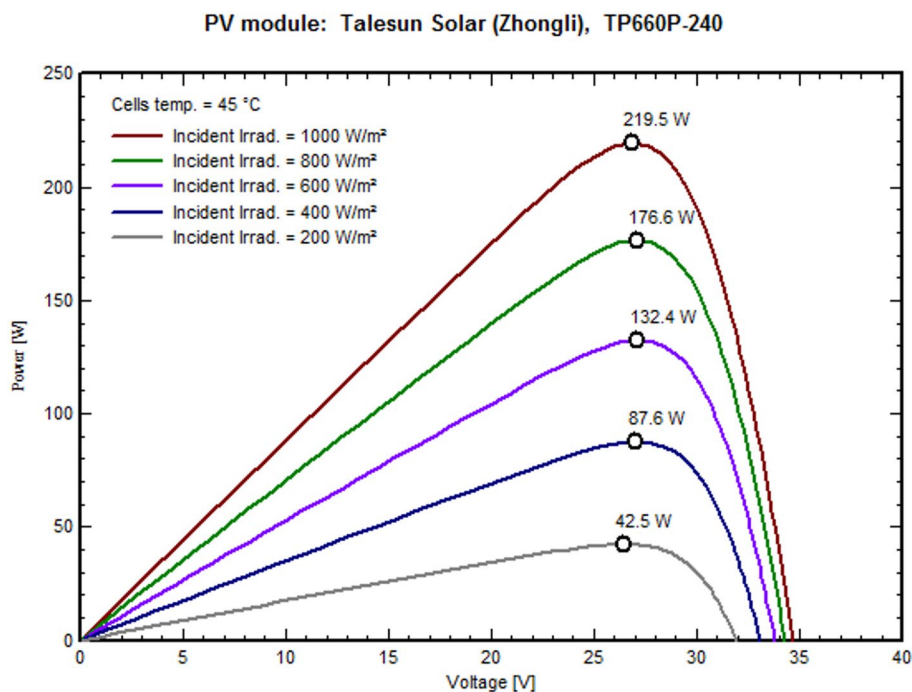


Fig. 6 Power vs. Voltage (P–V) characteristics of PV module using PVsyst software

Figure 5 shows all major components of the PV–Diesel–Battery configuration, including the Peak Power Tracker (PPT), inverter, battery charging unit, array diode, and mode controller.

PV module and inverter performance simulation

The PV module performance was analyzed using PVsyst software to account for variations in irradiance and temperature.

Figures 6, 7, 8 and 9 collectively illustrate the PV module's performance characteristics. As irradiance decreases, both maximum power and current output decline, while voltage remains relatively stable. Efficiency drops with rising cell temperature—from 16% at 10 °C to about 11% at 70 °C—emphasizing the importance of module ventilation.

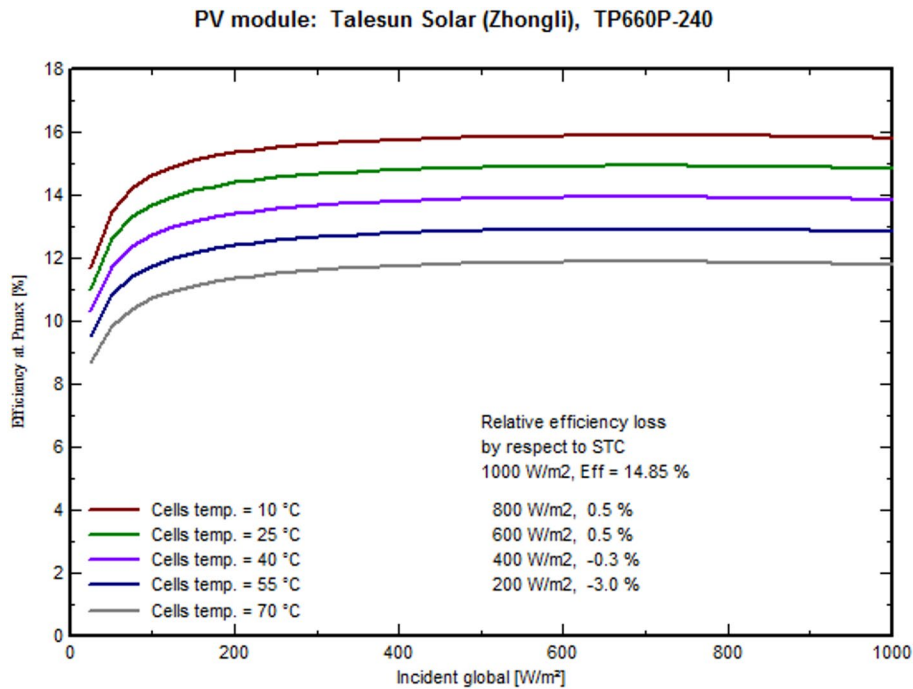


Fig. 7 Efficiency vs. incident irradiance of PV module using PVsyst software

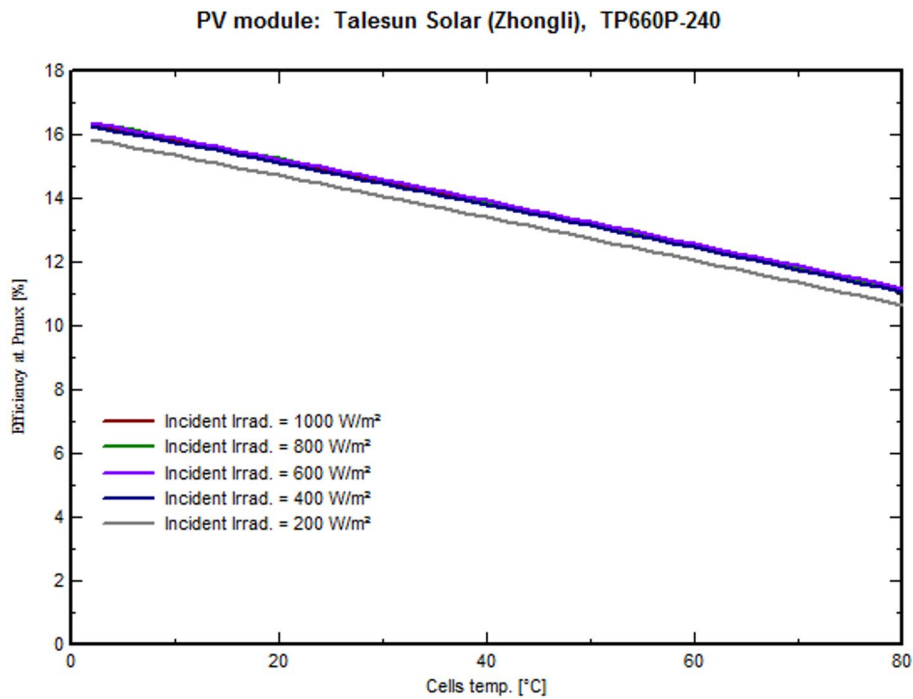


Fig. 8 Efficiency vs. cell temperature of PV module using PVsyst software

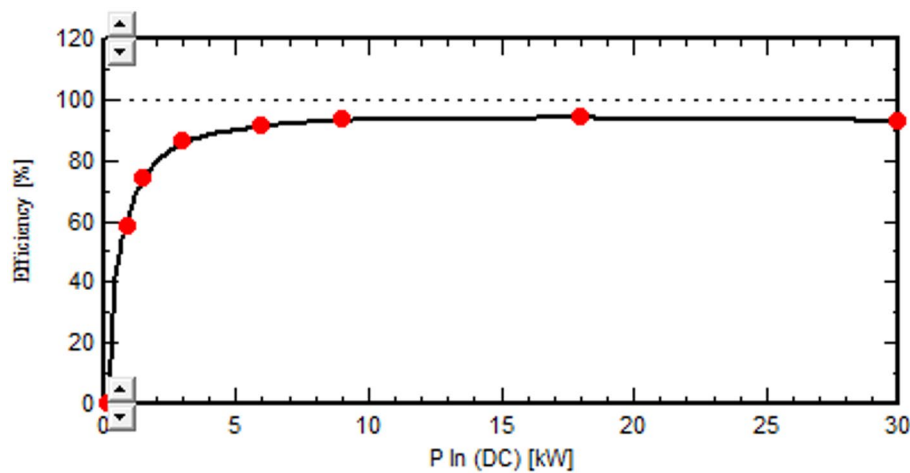


Fig. 9 Efficiency vs. input power of inverter using PVsyst software

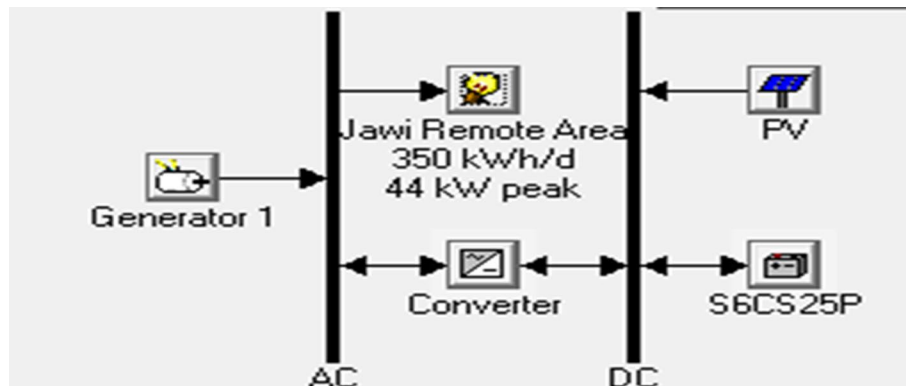


Fig. 10 Schematic diagram of the HOMER hybrid power supply system

The inverter achieves peak efficiency (~98%) in the 5–20 kW range, confirming that system sizing was appropriate.

Modeling, Simulation, and optimization

The hybrid configuration was modeled using HOMER software, incorporating an average daily load of 350.22 kWh. HOMER conducted:

- Simulation – hourly performance modeling for one year,
- Optimization – cost minimization and configuration ranking, and.
- Sensitivity Analysis – examining the impact of diesel and PV cost variations.

Figure 10 shows the HOMER model setup, including the PV array, diesel generator, converter, and battery storage units, along with their interconnections and load representation.

Load assessment and profiling approach

The daily energy demand of 350.22 kWh was derived from a structured load assessment process based on household surveys, appliance-level estimation, and community facility audits. A sample of households was surveyed to record the number, type, and usage duration of electrical appliances. The total daily energy consumption was calculated using standard aggregation formulas, $E = \sum(P_i \times t_i)$ and converted into hourly profiles

using diversity factors and typical usage schedules for lighting, mobile charging, refrigeration, small machinery, and institutional loads [25]. Load diversity considerations were incorporated to avoid overestimation of coincident demand, and the resulting peak-to-average ratio (1.52) reflects realistic operating conditions for rural communities. Seasonal variations—for example, increased lighting duration during rainy months and reduced daytime appliance use during agricultural seasons—were included to obtain a representative annual load profile used in HOMER simulations.

Results and discussion

The following results are presented concisely to highlight the system's technical, economic, and environmental performance. Each finding is followed by a short interpretation explaining the system behavior and the conclusions that can be drawn from the simulation outputs.

Solar resource analysis

The solar energy potential of the site was analyzed using HOMER software, with monthly solar resource data shown in Fig. 11. The figure also displays the clearness index, which represents the ratio of solar radiation reaching the Earth's surface to that at the top of the atmosphere. Values range from 0.45 for very cloudy months to 0.85 for sunny months. After entering the solar data, HOMER was run repeatedly, varying key parameters listed in Table 1 to find the most optimal system configurations. Sensitivity analysis was also performed by testing different diesel prices and PV module costs. The diesel price was 0.66 USD/L (local market rate) and PV prices were obtained from both global sources and the local market (40 BIRR/W).

System optimization

The simulation output included all feasible combinations of PV, generator, converter, and battery. Optimization results were given in two formats:

1. Overall list – all configurations ranked by Net Present Cost (NPC).

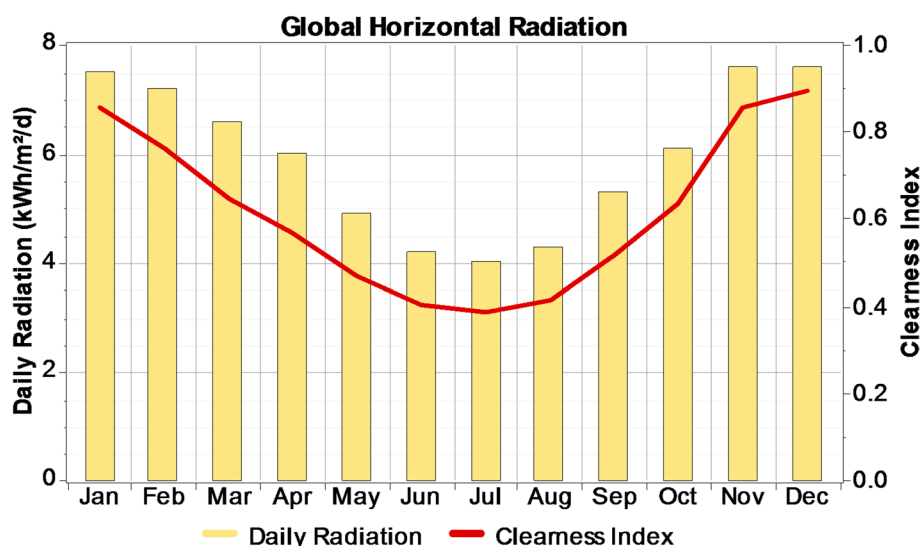


Fig. 11 Monthly average solar resources (irradiation and clearness index)

Table 1 Categorized optimization results

PV (KW)	Label (kW)	S6CS25P	Converter (kW)	Initial capital (Birr)	Operating cost (Birr/yr)	Total NPC (Birr)	COE (Birr/kW)	Renewable fraction	Diesel (L)	Label (hrs)
81	25	200	45	4,252,185	237,190	7,284,233	4.468	0.84	9377	1206
	20	40	20	298,408	864,633	11,342,748	6.93	0.00	46,622	7140
81	44		35	3,533,570	864,604	14,586,188	8.91	0.61	43,602	5754
	44			176,104	1,282,917	16,576,097	10.15	0.00	67,382	8759

2. Categorized list – only the least-cost configuration for each system type.

The most cost-effective system was a PV–generator–battery–converter setup with the generator operating under a Load Following (LF) strategy—producing only enough power to meet the primary load, leaving battery charging to renewable sources.

These results confirm that a hybrid PV–diesel–battery system can significantly reduce fuel use and cost while maximizing renewable energy contribution.

From the results in Table 1, the most cost-effective system is a PV–generator–battery–converter setup, with the generator running under a Load Following (LF) strategy. In this approach, the generator produces only enough power to meet the immediate primary load, while tasks like charging the batteries are handled mainly by renewable sources. For this configuration: Total Net Present Cost (NPC): 7,284,233 BIRR. Cost of Energy (COE): 4.468 BIRR/kWh. Renewable Energy Contribution: 84%. Diesel Consumption: 9,377 L per year. Generator Operating Hours: 1,206 h per year. This setup minimizes fuel use and operational costs while ensuring reliable power supply, making it the most efficient and economical option among all configurations tested.

In this hybrid system, renewable resources—mainly solar PV—provide a major share of the electricity supply, contributing 84% of the total annual energy. This high renewable fraction makes the setup both environmentally friendly and cost-effective, reducing dependency on diesel fuel. Figure 12 illustrates the monthly average electric production. Solar PV (yellow bars) consistently generates the majority of the electricity throughout the year. Diesel generator output (black bars) increases during months with lower solar production, such as May to September, when sunlight is reduced. In high-sun months like January, February, November, and December, the PV system alone meets nearly all the demand, and generator use is minimal or zero.

Comparison with existing studies

To validate the usefulness of the proposed system, the results were compared against recent hybrid energy studies (2020–2025) conducted in Sub-Saharan Africa and other developing regions. The optimized configuration in this study achieved a COE of 4.468 BIRR/kWh and an 84% renewable fraction. These values fall within the performance ranges reported in comparable hybrid PV–Diesel–Battery studies, which typically achieve 0.15–0.45 USD/kWh and 65–90% renewable penetration. Our significantly lower diesel consumption (9,377 L/year) results from optimized PV sizing and the Load-Following generator strategy. Additionally, the combined use of HOMER and PVsyst provides a more accurate evaluation of PV losses and inverter performance than many existing studies that rely on a single modeling tool. This comparison highlights the novelty and practical relevance of the proposed system for rural Ethiopian conditions.

Comparative discussion

The results of the proposed PV–Diesel–Battery system were compared with strategies reported in recent literature. Unlike many studies that rely only on HOMER optimization, this work combines HOMER with PVsyst, allowing more accurate estimation of PV performance under Ethiopian climatic conditions. Recent hybrid system studies typically report renewable fractions of 65–90% and COE values of 0.15–0.45 USD/kWh. The present system achieves an 84% renewable fraction and a COE of 4.468 BIRR/kWh (~0.15 USD/kWh), showing comparable or better performance. The reduced diesel

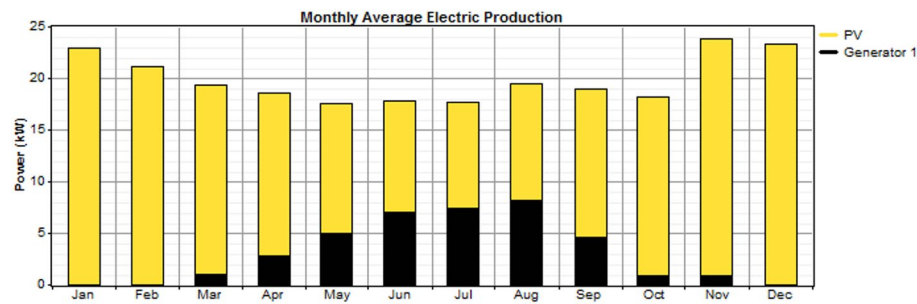


Fig. 12 Monthly Average Electric Production of the selected hybrid system

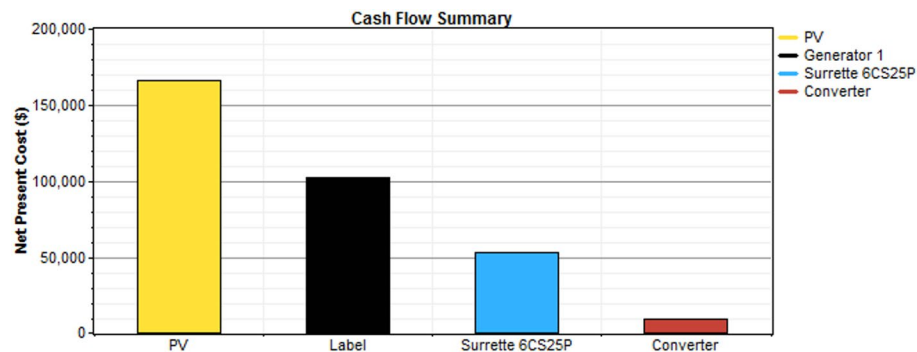


Fig. 13 Cash flow summary of the selected hybrid system

consumption is due to optimized PV sizing and the Load-Following strategy, which is more fuel-efficient than conventional generator control methods. Overall, the comparison shows that the proposed approach aligns well with existing strategies while providing improved contextual suitability for rural Ethiopia.

Economic and environmental assessment

This balance between PV and generator use ensures a reliable power supply while minimizing fuel consumption and operational costs. The system is well-suited for implementation in areas with strong solar potential but still requires backup generation during low-sun periods. The study identified several feasible system configurations, each with different levels of renewable energy contribution. The final choice depends on priorities—whether to minimize the initial investment cost or to maximize the long-term benefits of renewable energy use.

To determine the most suitable option, the HOMER model was run multiple times, testing different combinations of PV panel sizes, diesel generator capacities, battery banks, and converters. This process helped identify the setup that offers the best balance between performance and cost.

A sensitivity analysis was also carried out, changing key factors like diesel fuel price and PV panel cost to see how they affect the system's economics.

The Cash Flow Summary Fig. 13 shows that PV panels account for the largest share of the system's total cost, followed by the generator, batteries, and then the converter. This highlights the importance of optimizing PV sizing to balance upfront costs with renewable energy benefits. The following Table 2 indicates simulation results for different cases.

Table 2 Comparison of net present cost, energy cost and greenhouse gas emission diesel-only option with hybrid system

Component	Net present cost (Birr)	Cost of energy Birr/kwh	Diesel consumed per year (litter)	CO ₂ emission per year
Diesel only	16,576,097	10.15	67,382	177,439
PV/Diesel/battery	7,284,233	4.468	9377	24,693
Diesel/battery	11,342,748	6.93	46,622	122,770
Pv/diesel	864,604	8.91	43,602	114,818

Optimization using HOMER identified the PV–Diesel–Battery–Converter configuration (81 kW PV, 25 kW diesel, 45 kW converter, 200 batteries) as the most cost-effective system. It achieved: Net Present Cost (NPC): 7,284,233 Birr, Cost of Energy (COE): 4.468 Birr/kWh, Renewable Fraction: 84%, Diesel Consumption: 9,377 L/year, Generator Operating Hours: 1,206 h/year. The hybrid PV–Diesel–Battery system reduced CO₂ emissions by 86% and fuel consumption by nearly 58,000 L per year compared to the diesel-only configuration. These results align with previous research validating the hybrid model's technical and environmental viability.

The proposed hybrid PV–diesel–battery design directly benefits rural households, community institutions (such as schools and health centers), and small enterprises that lack reliable grid access. The optimized system configuration and sizing methodology provided in this study can serve as a practical guideline for local technicians, energy planners, and rural electrification agencies when designing and implementing similar off-grid systems in their daily operations.

In practical deployment, several constraints must be considered, including the high initial investment for PV modules and batteries, the need for skilled technicians for maintenance, and environmental factors such as dust accumulation that may reduce system performance. Additionally, fuel storage requirements, battery lifespan considerations, and the economic capacity of rural communities may influence system feasibility. These factors should be accounted for during planning and implementation.

Reliability and battery performance

Figure 14 shows two key performance indicators for the battery system. State of Charge (SOC) of batteries shown in green. On average, the batteries are 72.1% full over the year. From January to April, and again from October to December, the SOC stays high (close to or above 90%), meaning there is enough solar energy to keep the batteries well charged. Between May and September, SOC drops, reaching its lowest point in July and August (around 20–25%), due to reduced solar energy availability and higher demand.

Probability of Loss of Load (LOL) shown in brown. This represents the percentage of time the system cannot meet the energy demand because the batteries are empty. The yearly average LOL is 5%, which is relatively low. Most of this loss occurs in June, July, and August, when battery SOC is lowest. In months with high SOC, the LOL is nearly zero.

The system performs well for most of the year, with high battery charge levels and minimal risk of power shortage. However, in the middle of the year (especially July and August), the low battery charge increases the chance of not meeting the load, indicating the need for additional storage capacity or backup power during these months.

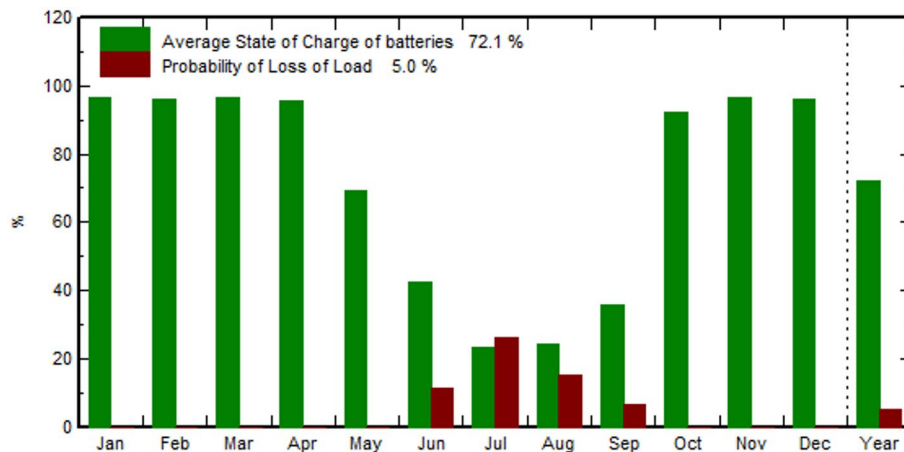


Fig. 14 Average state of charge (SOC) of batteries and probability of loss of load (LOL)

Sensitivity analysis

The hybrid system's robustness was verified by testing varying diesel prices (0.66–0.9 USD/L) and PV costs (40 Birr/W $\pm$ 20%). NPC increased slightly with higher fuel costs but decreased with falling PV prices. The hybrid configuration consistently remained the least-cost option under all tested conditions.

To strengthen the interpretation of the results, the discussion also reflects on how the proposed system-design approach addresses the practical challenges typically encountered in hybrid renewable energy planning. By emphasizing the system's performance behaviour, reliability, and cost trends under varying solar and fuel conditions, the discussion highlights the specific advantages of the adopted strategy—namely reduced diesel dependence, improved renewable penetration and realistic performance estimation using a combined techno-economic and PV modelling approach. Presenting the results in this manner clarifies the practical value of the proposed system and demonstrates how the adopted method provides a robust and applicable solution for rural electrification needs.

Conclusion

This study successfully designed, simulated, and optimized a hybrid photovoltaic–diesel–battery power system to meet the electricity demand of the remote Jawi (Buni Jirra) area in Ethiopia. The proposed configuration, developed using HOMER and PVsyst software, effectively addresses the challenges of off-grid rural electrification by combining renewable and conventional energy sources. The optimal system, comprising an 81 kW PV array, a 25 kW diesel generator, a 45 kW converter, and 200 batteries, achieved a renewable energy contribution of 84% with a net present cost of 7.28 million Birr and a cost of energy of 4.468 Birr/kWh. The hybrid configuration significantly reduced fuel dependency, limiting annual diesel consumption to 9,377 L and cutting CO₂ emissions by about 86% compared to a diesel-only system. The battery subsystem maintained an average state of charge of 72.1% and a loss of load probability of only 5%, ensuring high system reliability throughout the year. Sensitivity analyses confirmed the economic robustness of the hybrid model under varying fuel prices and PV module costs, emphasizing its long-term sustainability. Overall, the results demonstrate that hybrid PV–diesel–battery systems are not only technically feasible but strategically essential for rural

regions like Jawi where grid extension is impractical. Beyond the numerical optimization, the findings highlight a broader implication: decentralized hybrid systems can substantially improve energy security, reduce long-term dependence on fossil fuels, and support socio-economic activities in remote communities. By showing that high renewable penetration can be achieved at competitive cost, this study confirms the practical necessity outlined in the Introduction and provides an evidence-based framework that policymakers and rural electrification programs can directly apply when planning sustainable off-grid systems. These insights position hybrid renewable systems as a transformative pathway for rural development rather than merely an engineering solution.

Future research may focus on integrating additional renewable sources such as wind or biomass, implementing intelligent control algorithms for real-time energy management, and validating the system performance through experimental field studies. The outcomes of this study serve as a valuable reference framework for policymakers, engineers, and researchers working toward scalable, low-carbon energy solutions that enhance rural electrification and contribute to Ethiopia's sustainable development goals.

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Author contributions

YTM: Conceived and designed the experiments, EAM: Performed the experiment, HLA: Manuscript preparation, SJA: Contributed materials/analysis tools, MVS: Analyzed the data. All authors reviewed the manuscript. All authors were involved in editing the final manuscript. The authors read and approved the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Declaration of generative AI

Generative AI and AI-assisted technologies used only for formatting the content, grammar checking to improve the readability and language of the manuscript.

Competing interests

The authors declare no competing interests.

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